Project Proposal

Enterprise AI Copilot: On-Premise Natural Language to SQL Assistant

PREPARED BY

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TREND EGYPT FOR SYSTEMS

EXECUTIVE SUMMARY

Employees and clients using complex enterprise software often face a critical bottleneck: extracting specific, actionable data requires navigating convoluted screens or relying on technical teams for custom database queries. This results in wasted hours and operational inefficiency. This project program proposes the development of an intelligent, On-Premise AI Copilot deeply integrated into an existing enterprise architecture. By participating in this intensive project, senior engineering students will build a highly secure, privacy-first solution that translates natural language questions directly into accurate system insights, operating entirely on local servers to guarantee that no sensitive corporate data ever leaves the local environment.

Squad Composition & Tech Stack

We are assembling a cross-functional engineering squad consisting of senior Computer Science and Software Engineering undergraduates. The technology stack reflects modern enterprise architecture standards.

- **AI & Machine Learning (2 Students):** Focus on configuring and prompting Local LLMs, implementing secure Text-to-SQL pipelines, and establishing semantic understanding without external cloud dependencies.
- **Backend Engineering (2 Students):** Focus on .NET Core for building robust orchestration APIs, managing SQL Server database connections, and enforcing rigorous Row-Level Security (RLS) to ensure users only access authorized data.
- **Frontend Development (1 Student):** Focus on React to build a seamless, embedded AI chat interface within the primary enterprise system.

- **Project Lead / QA (1 Student):** Scrum master responsible for system testing, edge-case handling (e.g., SQL syntax errors), and maintaining alignment with business requirements.

Core System Modules (Development Milestones)

Module A: The Intent & Text-to-SQL Engine

The analytical core of the system. Students will deploy a local LLM environment and provide it with a customized, simplified schema (Semantic View Layer). The engine will take natural language inputs (e.g., "Show me the latest updates for this client") and translate them into secure, read-only SQL queries.

Module B: Secure Data Orchestrator

Built in .NET, this backend module acts as the strict gateway between the AI and the SQL Server. It intercepts the generated SQL, applies authentication and Row-Level Security filters, executes the query, and retrieves the raw JSON data. It also includes a self-correction loop to catch and repair faulty SQL syntax automatically.

Module C: Dynamic Reporting Workspace

This module manages the user experience. The .NET backend sends the retrieved raw database records back to the LLM with strict instructions to generate a clean, human-readable summary. The React frontend then displays this final, polished report seamlessly within the enterprise portal, turning raw rows into conversational insights.

Project Roadmap & Delivery Schedule

The project is structured as an agile delivery lifecycle over a full academic semester.

Phase	Duration	Key Objectives
Requirements & Architecture Setup	Weeks 1-3	Defining the scope, designing the database views, and setting up the local LLM and .NET backend infrastructure.
Core Integration & AI Pipeline Build	Weeks 4-8	Developing the Text-to-SQL logic, implementing .NET Row-Level Security, and connecting the React interface.
Testing & Self-Correction Refinement	Weeks 9-11	Rigorous testing of database edge cases, refining prompt engineering to prevent hallucinations, and handling query errors.
Final Polish & Demo Day	Weeks 12-14	System documentation, bug fixing, and live demonstration of the Copilot to university faculty and technical evaluators.

Strategic Value & Future Opportunities

By completing this project, the squad delivers a highly marketable software architecture that solves real enterprise privacy concerns regarding AI adoption. For the students, this serves as a premier portfolio piece demonstrating mastery over cutting-edge full-stack development and enterprise-grade AI integration, potentially leading to immediate technical roles within the software development sector.